

Effect of Two Different Molting Procedures on a *Salmonella enteritidis* Infection

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ABSTRACT Previous studies have shown that inducing a molt using feed removal exacerbated an intestinal infection by *Salmonella enteritidis* (SE). The current study was conducted to determine whether inducing a molt using a molt diet would still cause a pause in egg laying but not exacerbate an intestinal SE infection. In Experiments 1 and 2, hens were either provided *ad libitum* access to layer feed (control), fed 45 g molt diet (molt-feed) daily, or deprived of feed for 14 d (molted), and were orally infected with 1×10^7 SE on Day 4 of molt. Egg lay ceased in hens subjected to both molt treatments. The percentage of hens shedding SE did not differ among treatment groups in Experiment 1, whereas in Experiment 2 the molted hens had significantly higher shed rates than the controls on Days 10, 17, and 24 postinfection and the molt-feed hens on Days 17 and 24 postinfection. Compared with both fed groups of hens, the molted hens shed significantly more SE in Experiment 1 on Day 10 postinfection, and in Experiment 2 the molted hens shed significantly more SE on all 4 sampling days. In Experiment 3, subgroups of hens within each treatment group received serial 10-fold dilutions of SE and intestinal shedding of the organism in each subgroup was determined 7 d later. The 50% infectious dose (ID_{50}) was calculated for each treatment group from these shedding results. The ID_{50} was 2.7×10^3 SE, 5.2×10^2 SE, and 1.3 SE for control, molt-feed, and molted hens, respectively, indicating that feed removal substantially increased the susceptibility of hens to an SE infection and the molt diet decreased this susceptibility. Little difference was observed in the pH of alimentary samples or of cecal contents from hens in each treatment group, indicating that increased severity of SE infection was not due to alterations of intestinal pH. Histologically, the molted hens exhibited more extensive inflammation of the intestinal tract at Day 4 postinfection compared with the unmolted group. Intestinal inflammation in the molt-feed hens was intermediate between the two. These results indicated that molt induction, using a molt diet, will not put hens at risk for the severe intestinal infection observed in birds subjected to feed removal.

(Key words: molt, stress, feed removal, *Salmonella enteritidis*, intestinal disease)

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INTRODUCTION

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Induced molting is commonly used by the layer industry in the United States to

stimulate multiple egg-laying cycles in hens. It was estimated in 1987 that approximately 60% of hens nationwide and almost 90% of hens in California were molted (Bell, 1987). Feed removal remains the primary means of achieving the egg-laying pause (Brake, 1993). Recent research, however, has shown that induced molting through feed removal depressed cell-mediated immunity (Holt, 1992a) and decreased peripheral blood CT4⁺ (helper T cell subset) lymphocytes (Holt, 1992b). Hens infected with *Salmonella enteritidis* during the feed removal period had more severe intestinal infections, as indicated by significantly more extensive intestinal inflammation (Holt and Porter, 1992a; Porter and Holt, 1993) and significantly more *S. enteritidis* shed from the alimentary tract (Holt and Porter, 1992a,b). Increased intestinal shedding was observed whether the hens were infected concomitantly with the feed removal (Holt and Porter, 1992a,b) or 21 d prior to the feed removal (Holt and Porter, 1993).

As flock recycling remains an important part of the economics of the layer industry in the United States, methods alternative to feed removal to induce a molt were sought that might decrease the potential risk of *S. enteritidis* problems. Recently, Rolon *et al.* (1993) described a low-energy, low-density, low-Ca diet which, when given in limited amounts, induced molting as effectively as long-term feed removal. Because the hens remain on feed throughout the entire molt procedure, the physiological impact of administering such a molt diet may be less traumatic than total feed removal, and the effect on *S. enteritidis* infection may also be lessened. Therefore, a study was conducted that compared molt induction via feed removal vs special feed and their respective effects on intestinal *S. enteritidis* infection.

MATERIALS AND METHODS

Chickens

Birds used in the experiments were Single Comb White Leghorn hens retired from the laboratory's specific-pathogen-free production flock. The hens were housed in individual layer cages in a totally light- and environment-controlled building and, until the commencement of the molt procedure, were maintained on a 16 h light (L):8 h dark (D) regimen (lights on at 0100 h). Unless otherwise specified, the hens were allowed *ad libitum* access to water and antibiotic-free pelleted layer-breeder feed (16.7% CP, 2,974 kcal ME/kg, 3.0% Ca, .44% P).

Infection

Nalidixic acid-resistant *S. enteritidis* Strain SE89-83122 (SE) is a phage Type 13 primary poultry isolate.³ The nalidixic acid-resistant mutant used in this study was derived by A. Hinton and was maintained as a frozen stock at -20 C. Three days prior to infection, one of the stocks was thawed and subcultured for 2 consecutive d on nutrient agar⁴ to eliminate any bacteria that were damaged from the freeze procedure. A 10-mL tube of tryptic soy broth was inoculated from the second plate. Birds in Experiments 1 and 2 received an oral dose by gavage of 1 mL of a 1:100 dilution of the overnight broth culture of the SE (1×10^7 bacteria as determined by dilution analysis). Birds in Experiment 3 received different doses of SE as described below. At the time of infection chickens of Experiments 1, 2, and 3 were 67, 72, and 58 wk of age, respectively.

Experiment 1. Effect of the Two Molting Protocols on *Salmonella enteritidis* shedding

Twenty-four hens were divided equally into three experimental groups, and allowed to acclimate for 21 d prior to induction of molt. The molted group of hens were induced to molt using a modification of a procedure by Brake *et al.* (1982). Seven days before feed deprivation, the chickens were exposed to 8L:16D, which

²Received from Arthur Hinton, Department of Botany and Microbiology, Auburn University, Auburn, AL 36849.

³Originally obtained from the National Veterinary Services Laboratory, Ames, IA 50010.

⁴All bacteriological media were obtained from Difco Laboratories, Detroit, MI 48232.

continued for the remainder of the experiment. Feed was removed on Day 1 and the birds remained off feed for 14 d. Following the period of feed deprivation, the hens were fed pullet-grower ration (15% CP, 3,000 kcal ME/kg, .8% Ca, .32% P) for the remainder of the experiment. The molt-feed hens were fed 45 g of the molt diet (15.8% CP, 2,150 kcal ME/kg, .94% Ca, .23% P) daily in special feeders located within each cage, which commenced on Day 1 and continued to the conclusion of the experiment. The control group was allowed *ad libitum* access to the layer-breeder diet throughout the experiment. On Day 4, all birds were infected orally with SE and sampled for SE shedding 3, 10, 17, and 24 d postinfection as described below.

Experiment 2. Effect of the Two Molting Protocols on Salmonella enteritidis Shedding and on Intestinal Inflammation due to Salmonella enteritidis

Forty-two hens were divided into three experimental groups of 14 birds each and subjected to premolt procedures as outlined above. Birds in the three groups were infected orally with SE on Day 4 of molt. Eight of the birds in each group were repeatedly sampled for SE shedding 3, 10, 17, and 24 d postinfection as described below. Four days postinfection (Day 10 of molt), the six birds in each group not sampled for SE shedding were euthanized by cervical dislocation and segments of cecum, cecal tonsil, and colon were immersed in 5% neutral buffered formalin. Portions of each intestinal segment were then embedded in paraffin, sectioned at 3 μ m, and stained with hematoxylin and eosin. Histologic sections of intestine from control, molted, and molt-feed infected chickens were evaluated in blind fashion, scored on the degree of inflammation, and numerically ranked as described previously (Porter and Holt, 1993). The number of sections evaluated for each segment of intestine were two from cecum (middle and

distal regions), one from cecal tonsil, and two from colon. The final score for each tissue was the sum of criteria 1, 2, and 3 (Porter and Holt, 1993) plus whether heterophils were present in the epithelial layer (yes = 1 and no = 0). Ten was the highest score and 0 was the lowest score. For each segment of intestine, the mean inflammation scores of the different treatment groups were calculated.

Experiment 3. Effect of the Two Molting Protocols on Susceptibility of Hens to Salmonella enteritidis

Seventy-one hens were divided into the three treatment groups as described above and subgroups within each group were orally infected with SE on Day 4 of the molt treatment according to the following protocol: control, six birds each received 10^{-4} or 10^{-6} dilution and five birds each received 10^{-3} or 10^{-5} dilution; molt-feed, six birds each received 10^{-4} or 10^{-6} dilution and five birds each received 10^{-3} , 10^{-5} , or 10^{-7} dilution; molted, six birds each received 10^{-7} or 10^{-9} dilution and five birds each received 10^{-6} or 10^{-8} dilution. The undiluted culture contained 1.3×10^9 SE/mL as determined by dilution analysis. Seven days postinfection all birds were sampled for SE shedding as described below.

Intestinal Shedding of Salmonella enteritidis

Chickens were assayed for intestinal shedding of SE using a modification of a described procedure (Holt and Porter, 1992c). Eighteen hours before each sample collection, feed was removed from all birds that were being fed, and on the test day each bird received an intraperitoneal injection of .5 mL of 5% pilocarpine⁵ to induce hypersecretion of the intestine. Intestinal shedding can be determined from the resulting secretions (Holt and Porter, 1992c). Immediately following pilocarpine administration, the alimentary secretions from each bird were collected for 1 h in 23 $\times$ 28 cm food-grade plastic foam trays⁶ placed under each individual layer cage.

In Experiments 1 and 2, the secretions were serially diluted (10-fold) in PBS, and 100 μ L of each dilution and the undiluted

⁵All chemicals and antibiotics were obtained from Sigma Chemical Co., St. Louis, MO 63178-9916.

⁶Genpack Corp., Glens Falls, NY 12801.

sample was spread onto brilliant green agar containing 20 μg novobiocin/mL and 50 μg nalidixic acid/mL (BGNN). One milliliter of each secretion sample was also transferred to 9 mL tetrathionate broth, and the plates and tetrathionate broths were incubated overnight at 37 C. The SE were enumerated and for any samples in which no SE was detected, 100 μL of the tetrathionate enrichment for that sample was spread onto BGNN and examined for the presence of SE following overnight incubation at 37 C. As the threshold of detection was one SE organism per milliliter, samples that had no growth on the primary culture plates and the tetrathionate enrichment were assigned 0 SE per milliliter. Samples that had no growth on the primary culture plates but were culture-positive following tetrathionate enrichment were presumed to contain 9 SE/mL.

In Experiment 3, 1 mL of each secretion was enriched in 9 mL tetrathionate broth overnight at 37 C and then 100 μL of the tetrathionate samples were spread onto BGNN. The plates were incubated overnight at 37 C and the plates exhibiting SE growth were then determined. The mean 50% infectious dose (ID_{50}) endpoints of SE for each treatment group were determined by the method of Reed and Muench (1938).

Effect of the Molting Procedures on Alimentary pH

In Experiment 2, following the culture of the secretions for SE, the pH of each alimentary secretion was determined each sample day using a Beckman $\Phi 34$ pH meter.⁷ In Experiment 3, 24 h after sampling for SE shedding, 13 birds from each treatment group receiving the lowest challenge doses of SE were killed by cervical dislocation. The cecum was removed from each bird, .2 g of cecal material was mixed with .8 mL distilled water, and pH was measured immediately.

Statistical Analysis

The numbers of SE per milliliter shed in each treatment group (molted vs molt-feed

vs control) were transformed to $\log_{10}$ and then means were calculated. Means were also calculated for the percentage SE shed rate, alimentary secretion pH, and cecal contents pH. Significant differences between treatment groups were calculated using one-way analysis of variance (Shott, 1990) for all variables except for the evaluation of the rank sums of intestinal inflammation, where significant differences were determined using the Mann-Whitney rank test (Shott, 1990).

RESULTS

Experiment 1

Egg lay by hens in both molt treatment groups ceased within 7 d of feed removal or commencement of the molt diet. This was observed in all three experiments. The SE shed rate by the three groups of hens is shown in Figure 1A. Although the percentage of hens that were SE culture positive was higher in the molted group, there was not a significant difference among the three treatment groups on the 4 sampling d. Two of eight (25%) of the molted hens had blood in their alimentary secretions on Day 3 postinfection whereas neither of the fed groups of birds exhibited any blood. The molted hens shed 2 logs more SE than the fed hens at Day 3 and 4 logs more SE ($P \leq .001$) at Day 10 postinfection (Figure 2A). The amount of SE shed by hens in all three treatment groups decreased thereafter such that by Day 17, the groups all shed approximately 1 log SE per milliliter of alimentary secretion.

Experiment 2

There was no significant difference in the shed rate by all three treatment groups at Day 3 postinfection (Figure 1B). However, by Day 10 postinfection, the shed rate of the control hens was significantly lower ($P \leq .05$) than the molted hens, and the molt-feed group was intermediate between the other two groups but not significantly different ($P \leq .06$) from the shed rate of the molted hens. By Day 17 postinfection, both groups of fed hens had a significantly lower shed rate ($P \leq .05$, $P \leq .01$ for the control group and molt-feed group, respectively), and these two

⁷Beckman Instruments, Inc., Fullerton, CA 92634-3100.

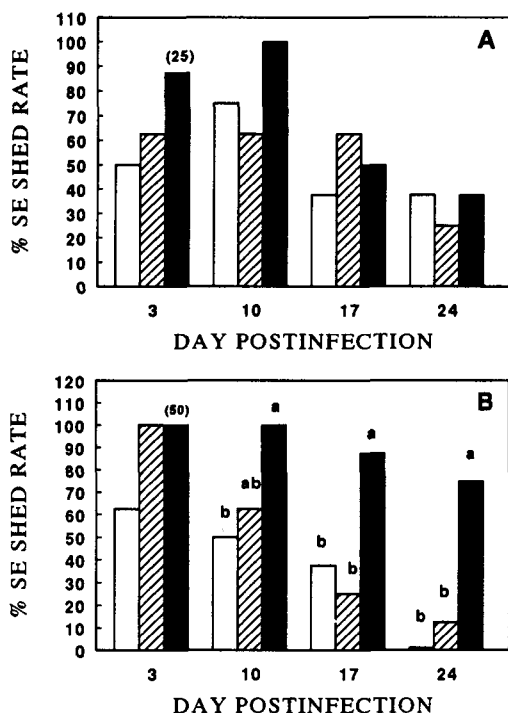


FIGURE 1. Effects of different molting procedures on *Salmonella enteritidis* (SE) shed rate in Experiment 1 (A) and Experiment 2 (B). Results represent the percentage of hens that were SE positive on the days sampled. Open bar = hens receiving *ad libitum* layer-breeder feed (control); lined bar = hens receiving 45 g special molt diet daily (molt-feed); filled bar = hens having feed removed for 14 d followed by pullet-grower diet for remainder of experiment (molted); $n = 8$ hens per group. Within days, bars with a different letter are significantly different ($P \leq .05$). The numbers in parentheses indicate the percentage of hens with blood in their alimentary secretions.

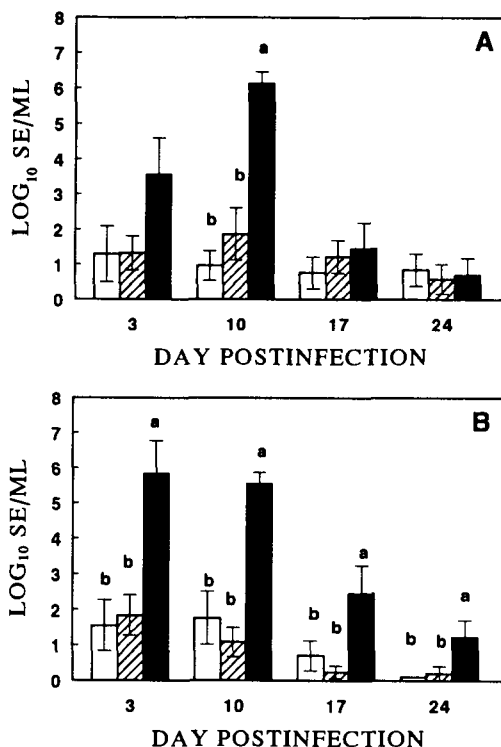


FIGURE 2. Effects of different molting procedures on *Salmonella enteritidis* (SE) numbers in Experiment 1 (A) and Experiment 2 (B). Results represent the $\log_{10}$ number SE/mL $\pm$ standard error of the mean of alimentary secretions on the days sampled. Open bar = hens receiving *ad libitum* layer-breeder feed (control); lined bar = hens receiving 45 g special molt diet daily (molt-feed); filled bar = hens having feed removed for 14 d followed by pullet-grower diet for remainder of experiment (molted); $n = 8$ hens per group. Within days, bars with a different letter are significantly different ($P \leq .05$).

groups continued to have a significantly lower shed rate at Day 24. Four out of eight molted hens (50%) had blood in their alimentary secretions on Day 3 postinfection, whereas no secretions from either of the fed groups of birds exhibited any blood. The molted hens shed significantly more SE, approximately 4 logs ($P \leq .001$), than either of the fed groups of hens on Day 3 (Figure 2B). The molted hens continued to shed significantly more SE at Day 10 postinfection (4 logs [$P \leq .005$ for both the control and molt-feed hens]), Day 17 (2 logs [$P \leq .05$ and $P \leq .01$ for control and molt-feed, respectively]), and Day 24 (1 log [$P \leq .05$ for control and molt-feed]). During this time frame the pH of the secretions in all

three treatment groups remained around neutrality (Figure 3) and, except for a significantly higher pH of the secretions from the molted and molt-feed groups on Day 17, there was little detectable group-to-group difference in secretion pH on the other days examined.

Intestinal inflammation due to SE was significantly higher in the molted group compared with the controls (Table 1). Of the tissues examined, the ceca from the molted hens exhibited the greatest degree of inflammation ($P \leq .001$) compared with the controls, and significant differences ($P \leq .05$) between these two treatment groups were also observed for the cecal tonsils and the

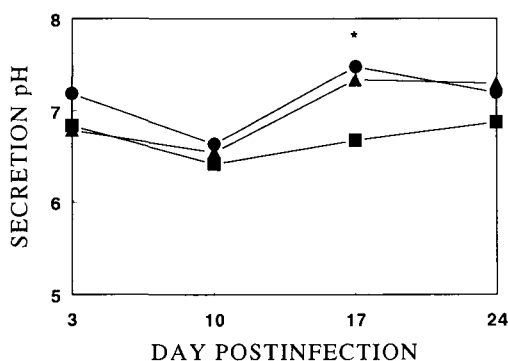


FIGURE 3. Effects of different molting procedures on alimentary secretion pH. Results represent the mean pH of alimentary secretions from the three groups of hens in Experiment 2. ■ = hens receiving *ad libitum* layer-breeder feed (control); ● = hens receiving 45 g special molt diet daily (molt-feed); ▲ = hens having feed removed for 14 d followed by pullet-grower diet for remainder of experiment (molted). The standard errors of the means were all less than 5%. $n = 8$ hens per group. The asterisk indicates that molted treatments were significantly different from control ($P \leq .05$).

colons. The level of inflammation observed in the tissues of the molt-feed hens was intermediate to that observed for the control and molted hens and was not significantly different from either treatment group.

Experiment 3

As the dose of SE administered to all three treatment groups of hens decreased, the percentage of SE-positive hens similarly

decreased (Figure 4). The calculated ID_{50} was 2.65×10^3 SE for the control hens, 5.17×10^2 SE for the molt-feed hens, and 1.3 SE for the molted hens. Two of five and three of six molted hens receiving the 1.3×10^3 SE dose and 1.3×10^2 SE dose, respectively, had blood in their alimentary secretions whereas one of the six molt-feed hens receiving the 1.3×10^5 SE dose contained blood. No alimentary samples from control hens contained blood. The pH of cecal contents from the hens taken 24 h later were determined to be $6.7 \pm .11$, $6.6 \pm .05$, and $6.8 \pm .06$ for the control, molt-feed, and molted groups, respectively.

DISCUSSION

In the current study, feed removal altered the course of an intestinal infection by SE. The amount of SE was significantly increased in these birds by 4 logs of SE as compared with controls (Figure 2), similar to previously observed results (Holt and Porter, 1992a,b). When birds were molted using the special feed, the amount of SE shed was decreased to levels comparable to those of the controls. Such results would be significant for the producer, as this would decrease the amount of SE released into the layer facility. Previous studies have shown that a certain number of organisms were required to initiate an infection—for SE approximately .6 to 5×10^4 organisms were necessary to infect a hen (Holt, 1993). By decreasing the amount of SE released into the layer facility environment, the chance of infect-

TABLE 1. Comparison of intestinal inflammation by *Salmonella enteritidis* in different molt treatment groups¹

Tissue	Treatment group ²		
	Control	Molt-feed	Molted
Colon	2.4 ± .78 ^b	2.9 ± 1.85 ^{ab}	6.67 ± 1.64 ^a
Cecal tonsil	1.17 ± .83 ^b	2.83 ± 1.64 ^{ab}	5.0 ± 1.46 ^a
Cecum	3.67 ± .73 ^b	6.33 ± 1.52 ^{ab}	9.33 ± .57 ^a

^{a,b}Mean ± SE within a row with no common superscript differ significantly ($P < .05$).

¹Numbers represent mean inflammation scores (± SE) of the three different intestinal tissues ($n = 6$ birds per group).

²Treatment groups: Control = hens receiving *ad libitum* layer-breeder feed; Molt-feed = hens receiving 45 g special molt diet daily; Molted = hens having feed removed for 14 d followed by pullet-grower diet for remainder of experiment.

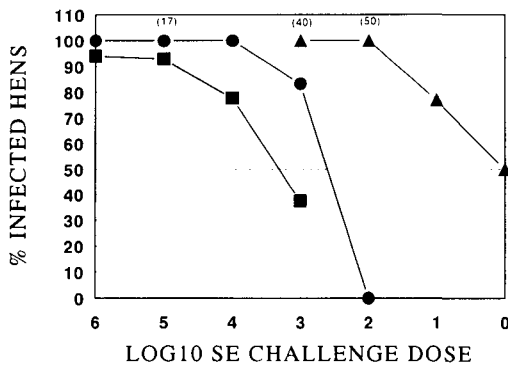


FIGURE 4. Comparison of the shed rates of hens subjected to different molting procedures 7 d after receiving different serial 10-fold dilutions of *Salmonella enteritidis* (SE). ■ = hens receiving *ad libitum* layer-breeder feed (control); ● = hens receiving 45 g special molt diet daily (molt-feed); ▲ = hens having feed removed for 14 d followed by pullet-grower diet for remainder of experiment (molted). The numbers in parentheses indicate the percentage of hens with blood in their alimentary secretions.

ing neighboring hens would also be diminished, and there would be less of the organism with which to contend during disinfection and clean up of a facility before the start of a new flock.

The SE shed rate was generally less affected by the 14-d feed removal procedure, similar to observations in previous studies (Holt and Porter, 1992b, 1993). The molt diet, therefore, had less of an ameliorating effect on the SE shed rate, although significant effects were observed in Experiment 2 (Figure 1B). Interestingly, although 100% of the hens on the molt diet in Experiment 2 were SE culture positive on Day 3 postinfection, the amount of SE shed by these hens was exceedingly low, 1 to 2 logs SE per milliliter of alimentary secretion, and this was approximately 4 logs less than the molted hens (which were also 100% culture positive). These results indicate that, in some cases, the shed rate only gives a partial representation of an SE problem within a flock and that an examination of other variables may provide a more complete picture.

An observation of blood in secretions from a certain proportion of the molted hens in Experiments 1, 2, and 3 and rarely in secretions from either of the fed groups

of birds indicated that the SE infection produced more intestinal inflammation in the molted birds. Histological examination of intestinal tissues from hens in the three treatment groups showed that the SE-infected molted hens did exhibit significantly greater intestinal inflammation as compared with the controls, especially in the cecum (Table 1). Similar observations regarding feed removal and its effects on intestinal infections have been made previously (Formal *et al.*, 1958; Holt and Porter, 1992a; Porter and Holt, 1993). Tissues from the molt-feed groups of hens exhibited a degree of inflammation intermediate between the molted hens and the controls. This inflammation was not different from either treatment group and indicates, as do the numbers of SE shed, that the molt diet did decrease the severity of the SE intestinal infection. However, the birds on the molt-feed diet did exhibit more intestinal inflammation than controls, which indicates that the molt diet did not eliminate all of the factors responsible for the observed exacerbation of the SE intestinal infection by induced molting.

The susceptibility to an SE intestinal infection was also altered in the molt-feed group as compared with the molted group. As was observed for the intestinal shedding and inflammation, the susceptibility to an SE infection was intermediate between that observed for the control and molted hens, which indicates that the molt-feed protocol does substantially reduce the susceptibility to an SE infection found in the molted hens. However, the hens on the molt feed still appear to be more susceptible than controls to an intestinal infection by SE, so the problem of increased susceptibility has not been completely resolved.

Because the molt diet does appear to reduce the severity of an intestinal infection by SE, it would be important to gain an understanding as to the reasons for this observed effect on the infection. The resident flora and the pH of the intestinal tract has been shown to be important in preventing infections by intestinal pathogens (Hentges and Freter, 1961; Bohnhoff and Miller, 1962; Bohnhoff *et al.*, 1964; Corrier *et al.*, 1990). The possibility therefore existed that the feed removal protocol

may have been altering the microecology of the intestinal tract, which could impact on the course of an intestinal SE infection. Such effects were observed previously for SE infection in mice treated with streptomycin (Bohnhoff and Miller, 1962; Bohnhoff *et al.*, 1964). The pH of the intestinal tract of the antibiotic-treated mice was increased and the concentrations of several volatile fatty acids were altered. This was correlated with an increased susceptibility to SE infection. Further, Corrier *et al.* (1990) have shown the protective effects of administered cecal flora and the consequent lowered cecal pH against the colonization of the intestinal tract of broiler chicks with *Salmonella typhimurium*.

In the current study, there was no detectable difference between treatment groups with regards to the pH of either alimentary secretions (Figure 3) or cecal contents. Although the pH of secretions from molted and molt-feed hens were the same, the amount of SE recovered from molt-feed hen secretions was significantly less than that found in molted hens and more closely resembled the amount of SE recovered from control hens (Figure 2). Therefore, little correlation between secretion pH and SE recovery was apparent. Alterations of intestinal pH may therefore not be responsible for the differences observed in SE infection severity, although differences in the volatile fatty acids found in the cecal contents of the treatment groups may still play a role in the course of the SE infection (D. Corrier, USDA, College Station, TX 77845, personal communication).

Changes in cell-mediated immunity and T cell subsets were observed previously in hens subjected to feed withdrawal (Holt, 1992a,b), and these observations prompted the studies examining the effects of induced molting on an SE infection. A possible explanation for the differences in the course of the SE infection may be their relative impacts on the immune system of the birds. The molt diet may potentially be less stressful and therefore exert less of an effect on the immune system of the hens compared with feed-deprived hens while having more of an effect compared with controls. These differential effects on im-

munity may then be reflected in the severity of SE infection and such effects need to be examined. Further, the relative "stressfulness" of each molt procedure also needs to be delineated using a panel of assays to measure stress such as corticosterone levels (Harvey *et al.*, 1984), heterophil to lymphocyte ratios (Gross and Siegel, 1983), or possibly leukocyte heat-shock proteins, recently shown to be induced during stressful situations (Wang and Edens, 1993). Whatever the cause, however, an understanding of the reasons for the decreased severity of infection by this organism in hens fed the molt diet is very important as this will help in the understanding of factors that are necessary for the protection against an SE infection. These factors might then be exploited in the field to increase resistance of hens to an infection by SE.

The current study indicates that inducing a molt through a special diet can decrease the severity of an infection by SE compared with molting via feed removal. Alterations of the current molt feed protocol may further decrease the progression of the SE infection as may other methods to induce a molt. Such protocols may allow the producer in the future to recycle their laying flocks while minimizing the potential for exacerbating an infection by SE or some other microbial agent (Swayne *et al.*, 1992).

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